

ABC STARTUP SOLUTIONS

I. Company Overview

- **Legal Name:** ABC STARUP SOLUTION
- **Office Location:** Suite 402, Innovate Tower, Cybertech Boulevard, Neo-Metro
- **Nature of Business:** Software Development & Cloud Applications

II. Mission & Vision

Company Mission

To engineer scalable renewable energy systems and intelligent energy management software that empower businesses and communities to cut emissions, lower energy expenditures, and accelerate the global energy transition.

Company Vision

To build a carbon-neutral planet where clean, resilient energy is universally accessible, intelligently managed, and seamlessly integrated into community infrastructure and industrial operations.

III. Organizational Structure

- **Managing Director / CEO:** Overall strategic leadership, investor relations, and company growth.
- **IT Lead / Senior Systems Architect:** Oversees 5 IT specialists (Developers, DevOps, SysAdmin).
- **Finance Manager:** Oversees 5 Finance personnel (Accounting, Compliance, Invoicing).
- **Sales & Marketing Lead:** Oversees 6 Sales reps and account executives.
- **HR & Operations Lead:** Oversees 4 HR & Office Administration specialists.

Enterprise Hardware Inventory

Tailored hardware inventory scaled specifically to meet the daily computing, networking, storage, and power requirements of all 20 employees across the 4 departments.

Asset ID	Hardware	Qty	Department	Purpose
AST-DSK-001	Desktop Computers (High-Performance Workstations)	9	IT (5) & Finance (4)	Heavy software compilation, IDEs, Docker containers, database management, and complex financial modeling spreadsheets.
AST-LAP-001	Business Laptops (Ultra-Portable)	11	Sales (6), HR (4), Finance Lead (1)	High-mobility use, on-site client presentations, recruitment interviews, and remote/hybrid work flexibility.
AST-MON-001	27-inch Dual-Display Monitors	25	All Departments	Provides dual-monitor setups (2 monitors)

				each) for the 9 desktop users and docking stations for mobile laptops.
AST-SRV-001	2U Rackmount Server	1	IT / Server Rack	Hosts local domain controller (Active Directory/LDAP), local DNS/DHCP, internal code staging repository, and local file shares.
AST-SWT-001	24-Port Managed Gigabit PoE+ Switch	2	IT / Network Infrastructure	Provides 48 total wired Gigabit ports for all 20 workstations, server, NAS, network printers, and PoE for wireless APs with VLAN support.

AST-RTR-001	Enterprise Gigabit VPN Router	1	IT / Network Infrastructure	Single-floor boundary firewall, secure remote VPN access for remote sales/employees, NAT, and traffic bandwidth management.
AST-WAP-001	Wi-Fi 6 Enterprise Wireless Access Points	2	Office Floor (Main Area & Conference)	High-density Wi-Fi 6 wireless coverage across the single floor with segmented SSIDs for Corporate, Voice, and Guest access.
AST-NAS-001	4-Bay Network Attached Storage(NAS)	1	IT / Centralized Storage	Centralized, high-speed on-premise local data backup, project repositories, shared departmental

				folders, and daily disk snapshots.
AST-EBD-001	Rugged External Encrypted Backup Drives (2TB)	3	IT / SysAdmin Cold Storage	Air-gapped offline weekly/monthly backups rotated off-site following the 3-2- 1 backup strategy for disaster recovery compliance.
AST-PRN-001	Multifunction Network Color Laser Printer	1	Shared Office Central Area	Shared departmental printing, document scanning to email/NAS, employee badge/PIN secure release, and contract printing.
AST-UPS-001	Smart Online Rackmount UPS (1500VA)	2	IT Server Rack & Network Enclosure	Protects the core server, switch, router, and NAS against

				sudden power surges and supplies emergency runtime for safe shutdowns.
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Hardware Selection & Quantity Justifications

1. End-User Compute (9 Desktops, 11 Laptops, 25 Monitors):

- Exactly matches the 20-employee headcount.
- The 5 IT engineers and 4 Finance specialists are assigned stationary workstations with dual displays for compute-heavy coding and financial sheets.
- The 6 Sales representatives, 4 HR specialists, and Finance Lead are assigned laptops for mobility during client pitches, candidate interviews, and meetings.

2. Network & Server Infrastructure (1 Server, 1 Router, 2 Switches, 2 WAPs):

- For a single-floor, 20-person startup, a single 2U enterprise server efficiently handles local network authentication (Active Directory), file sharing, and internal test staging.
- Two 24-Port PoE+ switches (48 total ports) provide ample connectivity for all 20 wired desks, network peripherals, access points, and future hires.
- Two Wi-Fi 6 WAPs deliver seamless roaming across the open workspace and private meeting rooms without dead spots.

3. Storage, Power & Continuity (1 NAS, 3 Backup Drives, 1 Printer, 2 UPS):

- A 4-Bay NAS provides fast, local on-premise shared storage.
- Three encrypted external drives implement an air-gapped 3-2-1 backup strategy to prevent data loss or ransomware risks.
- Two 1500VA UPS units protect the core server and network switches against power loss and voltage fluctuations.

Enterprise Software Inventory

Software	Version	License	Purpose
Windows 11 Pro	64-bit (23H2 / 24H2)	Commercial OEM / Volume (20 Seats)	Primary client OS providing BitLocker full-disk encryption, Windows Sandbox, Group Policy (GPO) integration, and domain joining.
Ubuntu Server	24.04 LTS	Open Source (GNU GPL)	Host OS for the on-premise rackmount server running local containerized staging environments, local DNS/DHCP, and automated backup daemons.
Microsoft Office 365	Business Standard	Commercial Subscription (20 Users)	Core office suite (Word, Excel, PowerPoint, Outlook, Teams) for inter-departmental collaboration, financial analysis, and client documentation.
Visual Studio Code (VS Code)	Latest Stable (v1.9x)	Open Source (MIT License)	Lightweight, extensible code editor and IDE for the IT development

			team with debugging, linting, and Git integration tools.
Git	v2.4x+	Open Source (GNU GPL v2)	Distributed version control system enabling developers to track source code changes, maintain feature branches, and manage code repositories.
GitHub Desktop	Latest Stable (v3.x)	Open Source (MIT License)	GUI client facilitating Git branch management, visual merge conflict resolution, commit history tracking, and repository synchronization.
Oracle VM VirtualBox	v7.0.x	Open Source (GNU GPL v3)	Type-2 desktop hypervisor allowing IT engineers and QA to spin up isolated virtual machines for multi-platform application testing.
Google Chrome	Enterprise Stable	Freeware / Proprietary (Enterprise MSI)	Standardized web browser with enterprise GPO

			policy controls, developer inspection tools, and access to cloud-based SaaS services.
Microsoft Defender for Endpoint	Integrated / M365 Security	Enterprise Commercial License	Real-time endpoint detection and response (EDR), behavioral antivirus, heuristics scanning, anti-malware, and network firewall protection.
AnyDesk	Standard Business	Commercial Named-User License	Encrypted, low-latency remote desktop management allowing the SysAdmin to troubleshoot workstations and assist mobile sales staff remotely.
7-Zip	v24.x	Open Source (GNU LGPL)	File archiving and compression utility supporting high-ratio compression and AES-256 encrypted file packaging for data transit.

Detailed Software Need & Justifications

- **Operating Systems (Windows 11 Pro & Ubuntu Server):**
 - Windows 11 Pro is required across all 20 end-user laptops and desktops to enforce BitLocker disk encryption (essential for mobile laptops carrying business data) and centralized Active Directory domain authentication.
 - Ubuntu Server 24.04 LTS provides a lightweight, stable, and cost-effective Linux environment for hosting internal web staging servers, databases, and network management services without incurring recurring server OS licensing fees.
- **Development & Version Control Toolchain (VS Code, Git, GitHub Desktop, VirtualBox):**
 - The 5 IT software developers and engineers require standardized development environments.
 - VS Code offers a uniform code editor supporting multiple programming stacks.
 - Git & GitHub Desktop provide essential version control and collaboration safety, ensuring code changes are tracked, reviewed, and backed up without overwriting work.
 - VirtualBox isolates testing environments, allowing engineers to test code across different OS versions safely without risking host workstation stability.
- **Productivity & Business Operations (Microsoft Office, Google Chrome, 7-Zip):**
 - Microsoft Office 365 provides industry-standard tools: Excel for Finance's financial modeling and payroll, Word for HR policies and client contracts, PowerPoint for Sales pitch decks, and Teams/Outlook for unified company communications.
 - Google Chrome Enterprise ensures uniform web rendering, secure enterprise extension policies, and fast access to cloud web applications.
 - 7-Zip allows staff to package large code builds, logs, and financial records with secure AES-256 encryption.

- **Security & System Administration (Microsoft Defender, AnyDesk):**
 - Microsoft Defender delivers continuous, zero-day threat protection across all endpoints, blocking ransomware and unauthorized scripts.
 - AnyDesk equips the Junior System Administrator with a secure, instant remote-access tool to diagnose issues, install updates, and assist remote Sales staff in the field without needing to physically travel to their location.

Enterprise Network Inventory

Equipment Name	Quantity	Location / Scope	Purpose & Specifications
ISP Modem	1	Demarcation Point / Server Room	Converts incoming ISP fiber-optic signal into a Gigabit Ethernet WAN feed for primary Internet connectivity.
Router	1	Main Equipment Rack (MDF)	Handles inter-VLAN routing, NAT translation, DHCP scopes, DNS forwarding, and secure VPN connections.
Firewall	1	Perimeter Boundary (MDF)	Enforces perimeter network security, deep packet inspection, intrusion prevention (IPS/IDS), and traffic filtering.

Switch	2	Main Distribution Frame (MDF)	24-Port Gigabit Managed PoE+ switches providing VLAN distribution and PoE power delivery to connected network devices.
Access Point	2	Main Floor & Conference Room	Dual-band Wi-Fi 6 enterprise access points providing secure, high-density wireless coverage and guest network isolation.
Patch Panel	2	MDF Server Rack	24-Port Cat6 1U rackmount panels providing structured termination and labeling for all horizontal in-wall cable runs.
CAT6 Cables	2 Spools + 50 Patch Leads	In-Wall Drops & Desk Connections	Solid copper bulk cabling for horizontal structural runs and pre-terminated patch cords for device connections.
RJ45 Connectors	100-Pack (1 Box)	Server Rack & Toolkits	High-grade 8P8C modular connectors used for terminating

			custom-length cable drops and on-site maintenance.
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Enterprise Network Diagram

System Administrative Roles

1. Helpdesk Technician

Responsibilities

- Provide technical support to employees and users.
- Troubleshoot hardware, software, and network issues.
- Install and update computer applications.
- Reset passwords and manage user accounts.
- Record and track support tickets.

Skills

- Customer service and communication
- Basic Windows and macOS troubleshooting
- Hardware and software installation
- Problem-solving
- Basic networking knowledge

Common Tools

- Microsoft Intune
- TeamViewer / AnyDesk
- ServiceNow or Jira Service Management
- Windows Device Manager
- Remote Desktop

Certifications

- CompTIA A+
- Google IT Support Professional Certificate
- ITIL 4 Foundation

2. Network Administrator

Responsibilities

- Install and maintain computer networks.
- Configure routers, switches, and firewalls.
- Monitor network performance and security.
- Troubleshoot connectivity problems.
- Manage Wi-Fi and VPN access.

Skills

- TCP/IP networking
- Routing and switching
- Network security
- Firewall configuration
- Network monitoring

Common Tools

- Cisco Packet Tracer
- Wireshark
- Cisco IOS
- pfSense
- SolarWinds

Certifications

- CompTIA Network+
- Cisco CCNA
- Juniper JNCIA

3. Linux System Administrator

Responsibilities

- Install and maintain Linux servers.
- Manage users, permissions, and file systems.
- Monitor server performance.
- Automate tasks using shell scripts.
- Perform system backups and updates.

Skills

- Linux command line
- Bash scripting
- System administration
- Server security
- Troubleshooting

Common Tools

- Ubuntu / CentOS / Debian
- Bash
- SSH
- systemd
- Cron

Certifications

- CompTIA Linux+
- Red Hat Certified System Administrator (RHCSA)
- LPIC-1

4. Cloud Administrator

Responsibilities

- Manage cloud infrastructure and virtual machines.
- Deploy applications and cloud services.
- Monitor cloud performance and costs.
- Configure storage, networking, and security.
- Ensure data backup and disaster recovery.

Skills

- Cloud computing
- Virtualization
- Identity and access management (IAM)
- Cloud security
- Automation

Common Tools

- Microsoft Azure
- Amazon Web Services (AWS)
- Google Cloud Platform (GCP)
- Docker
- Kubernetes

Certifications

- AWS Certified Cloud Practitioner
- Microsoft Azure Administrator (AZ-104)
- Google Associate Cloud Engineer

In one organization, the Helpdesk Technician assists employees with daily technical issues and escalates complex problems when needed. The Network Administrator ensures the company's network, internet, and security systems remain reliable. The Linux System Administrator manages the servers that host applications and company data, keeping them secure and

operational. Meanwhile, the Cloud Administrator maintains cloud services, storage, and virtual infrastructure. Together, they collaborate to provide a stable, secure, and efficient IT environment that supports the organization's daily operations.

Infrastructure Recommendations

1. Internet Provider

Recommendation: Enterprise Fiber Internet (1 Gbps) with a backup ISP.

Reason: The company relies on cloud applications, VPN access, Microsoft 365, and remote employees. A primary fiber connection plus a secondary backup ISP will ensure continuous operations if the main connection fails.

2. Server Specifications

Recommended Server Configuration

Component Specification

Server Type 2U Rackmount Enterprise Server

CPU Intel Xeon / AMD EPYC

RAM 64 GB ECC

Storage RAID SSD for system + NAS for shared files

OS Ubuntu Server 24.04 LTS

Reason: The server hosts Active Directory/LDAP, DNS, DHCP, internal code repositories, and file sharing, making it suitable for the company's development environment.

3. Backup Strategy

Recommendation: Implement the 3-2-1 Backup Strategy.

- 3 copies of critical data
- 2 different storage media (NAS + encrypted external drives)

- 1 off-site backup rotated weekly or monthly

Reason: This protects the business from ransomware, accidental deletion, hardware failure, and disaster recovery scenarios.

4. Security Recommendations

- Use an enterprise firewall with IPS/IDS.
- Separate Corporate, Voice, and Guest Wi-Fi using VLANs.
- Require VPN for remote access.
- Enable BitLocker encryption on all Windows devices.
- Apply regular security updates and centralized management.

Reason: These measures secure company data while allowing safe remote work and network segmentation.

5. Antivirus

Recommendation: Microsoft Defender for Endpoint.

Reason: It provides real-time antivirus, endpoint detection and response (EDR), ransomware protection, behavioral monitoring, and integrated firewall security for all employee devices.

6. Password Policy

Recommended Policy

Policy	Setting
Minimum length	12 characters
Complexity	Uppercase, lowercase, number, symbol
Multi-Factor Authentication Required	
Password reuse	Last 10 passwords blocked
Account lockout	After 5 failed attempts

Reason: Since the company uses Active Directory and VPN authentication, strong passwords and MFA reduce the risk of unauthorized access.

7. Expansion Plan

Recommendation for Future Growth

- Add another enterprise server for virtualization and redundancy.
- Upgrade from 48 to 96 network ports as employee numbers increase.
- Expand NAS storage from 4-bay to 8-bay.
- Deploy additional Wi-Fi 6 access points for larger office areas.
- Increase cloud backup capacity for growing project repositories.

Reason: The current infrastructure is designed for 20 employees, but its modular switches, enterprise networking, and centralized storage make it easy to scale as the company expands.

Overall Recommendation

A hybrid infrastructure combining enterprise fiber internet, a rackmount Ubuntu server, NAS storage, Microsoft Defender security, VPN access, and the 3-2-1 backup strategy will provide reliability, security, scalability, and business continuity for ABC Startup Solutions.

Personal Reflection

Working on the Enterprise Infrastructure Plan helped me understand how different parts of an IT infrastructure work together to support a company's daily operations. I learned that system administration is more than fixing computers it also involves planning hardware, designing secure networks, managing servers, implementing backup strategies, and protecting company data. Creating the hardware inventory, network infrastructure, software inventory, and administrative roles showed me how each component contributes to a reliable and efficient business environment.

The most challenging task was designing the enterprise infrastructure and deciding the appropriate hardware, networking equipment, and server specifications for a 20-employee company. It required balancing performance, cost, scalability, and security while ensuring that every department had the resources they needed. Understanding how devices such as managed switches, Wi-Fi 6 access points, NAS storage, VPN routers, and Ubuntu servers connect within one network was initially difficult, but it became clearer as I organized the infrastructure into logical sections.

Planning before deployment is extremely important because it prevents costly mistakes and reduces downtime. A well-planned infrastructure ensures that hardware is compatible, the network is properly segmented, security policies are established, and backup systems are ready before the company begins operations. Proper planning also allows future expansion, making it easier to add employees, devices, or cloud services without redesigning the entire system. In addition, documenting the infrastructure helps administrators maintain and troubleshoot the environment more efficiently.

This project will help me become a better System Administrator by strengthening both my technical knowledge and planning skills. I gained experience in selecting enterprise hardware, understanding Linux servers, implementing backup and security strategies, and recognizing the responsibilities of Helpdesk, Network, Linux, and Cloud Administrators. More importantly, I learned the value of documentation and structured decision-making, which are essential in real-world IT environments. As a BSIT student, this project gave me a practical view of how professional system administrators design, manage, and secure enterprise infrastructure, preparing me for future internships and careers in IT.

